Government of **Western Australia**
Department of **Health**

Psychotropic Medication:
Physical Health Monitoring Handbook
CAMHS

Department of
Health

GOVERNMENT OF
WESTERN AUSTRALIA

Child and Adolescent Mental Health Service (CAMHS)

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Introduction

The development of this handbook, physical health policies and accompanying forms initially occurred in response to the Office of the Chief Psychiatrist Thematic Review regarding physical care needs of mental health patients.

In the CAMHS setting, most psychotropic medication is prescribed 'off-label': outside the approved indications and manufacturers' recommendation. In this context, it is important to ensure that potential side effects and other associated health risks are closely monitored.

Identification of the physical health care needs of children and adolescents prescribed psychotropic medication should occur through the process of obtaining a medical history, performing a physical examination and conducting relevant investigations.

Findings should be documented in their medical record. This is mandated by the National Standards for Mental Health Services and the Physical Health Care of Mental Health Consumers Standard in the Chief Psychiatrist WA Standards of Clinical Care 2015

This handbook is not a comprehensive medical text or a substitute for sound clinical judgment or consultation with colleagues. While every care has been taken with accuracy, it should be noted that information can, and does, change rapidly. The Handbook provides monitoring tools for the metabolic side effects of psychotropic medication in children, based on international literature and existing protocols in Australia and overseas. It includes brief guidelines, easy-to-use forms, charts and a collection of reference materials to support clinicians. See Appendix 4 for a flowchart with links to monitoring charts, forms and screening tools.

General Points

(see Psychotropic medication - monitoring adverse physical health effects policy)

All assessment results must be communicated to other relevant care providers, including referral for further management of emerging condition. Responsibility for communicating assessment results lies with the primary prescriber or delegate, see CAMHS Physical Health GP Letter Template

All medication use and review must be fully documented and documentation should be easily accessible. The forms used for this purpose must be kept in front of the patient's chart or in a separate section of medical record.

Outcomes of medication reviews to be discussed at Multidisciplinary Team (MDT) clinical reviews if clinically indicated- see the CAMHS Multidisciplinary Team Clinical Reviews Policy.

Responsibility for monitoring patients receiving psychotropic medication lies with the primary prescriber of the medication, or a delegate. Any delegation of responsibility must be clearly documented.

Prescribing of medications where care is shared between CAMHS and outside services or alternates between acute, community and specialised settings within

CAMHS, should be decided following a discussion between a primary prescriber and other prescribers who are involved in providing care for the patient, see the CAMHS Transitions in Care Procedure.

Under section 304 of the MHA 20147 the provision of off-label treatment to any child who is an involuntary patient must be reported to the Chief Psychiatrist. The notification form found on the OCP site must be used and a copy of the form must be filed in the medical record.

CAMHS must ensure that young people and families agree with and are actively involved in the decision-making process and have adequate information about their management. This can be achieved through informed consent, which constitutes the process of obtaining patient's consent for a health practitioner to provide healthcare treatment. Medical record form Consent for Medication Use MR859.01/MR505.00 **must be used for all medication** prescribed in CAMHS as evidence of informed consent.

QTC prolongation risks

QTC prolongation risks are well recognized, though the change in the majority of cases is unlikely to embody significant clinical risks[1]. It is the responsibility of the physician/ psychiatrist to examine the risk-to-benefit ratio that accounts for the danger of sudden cardiac death before prescribing a psychotropic medication. Each individual patient should be assessed on the potential contributing factors to the prolongation of QTc interval, including evaluation of risk factors for sudden cardiac death. Clinical assessment and screening may also include probing and recording past medical history of syncope, exercise/exertion syncope and family history of sudden death.[1]

Present literature shows that many antipsychotics, regardless of the generations, some antidepressants and stimulants have the potential of causing QTc interval prolongation. QTc prolongation has a huge clinical importance as it is correlated with dangerous arrhythmias especially torsades de pointes (TdP) which may lead to sudden death. Patients with TdP may be asymptomatic but sometimes develop syncope and dizziness. It is recommended that there must an early collaboration with medical disciplines, e.g. cardiology referral, in order to adopt the optimal approach and to manage these risks. Shath et al (2014) recommend: 'it is not mandatory to perform ECG monitoring as a prerequisite in the absence of cardiac risk factors. An ECG should be performed if the initial evaluation suggests increased cardiac risk or if the antipsychotic to be prescribed has been established to have an increased risk of TdP and sudden death.'[1]

In relation to mood stabilising antiepileptic agents, the evidence suggests that 'while the preponderance of clinical data would suggest that use of most antiepileptic drugs does not pose excessive additional risk of QT prolongation, available data also do not provide sufficient evidence that these drugs are entirely free of risk in all patients. In particular, the potential for these medications, either alone or in

[1] Shah, A. a, Aftab, A., & Coverdale, J. (2014). QTc prolongation with antipsychotics: is routine ECG monitoring recommended? Journal of Psychiatric Practice, 20(3), 196–206.

combination, to prolong the QT interval should be considered.[2]'

Process Resources & Forms

Process	Resources & Forms
Before prescription	
Reasonable clinical judgment, supported by available assessment information, must guide the prescription of psychotropic medications. Baseline assessments and investigations prior to commencement of medication must be conducted for all children: • A physical examination documented on the CAMHS Physical Examination Form. This includes recorded measurements of BP, pulse, weight and height, Body Mass Index (BMI), waist circumference • Presence of abnormal involuntary movements including tics and extrapyramidal movements are to be noted on the physical examination • Recommended baseline investigations: FBP, UsE's, LFT's, B12, Folate and TFTs.	The CAMHS SSCD Physical Examination Form (printable and writeable). Abnormal Involuntary Movement Scale
Before prescription: additional requirements for specific medications	
Antipsychotic Medication The Antipsychotic Physical Health Monitoring Form has been developed to support clinicians in the safe use of Antipsychotic medication. It should be initiated prior to commencing Antipsychotic Medication. Baseline metabolic profile: fasting BSL, fasting insulin, cholesterol and triglycerides, HDL and LDL, Prolactin.	Antipsychotic Physical Health Monitoring Form. This includes risk factors identified for cardiac and metabolic disorders.

[2] Feldman AE, Gidal BE. QTc prolongation by antiepileptic drugs and the risk of torsade de pointes in patients with epilepsy. Epilepsy Behav. 2013 Mar;26(3):421-6.

Stimulant Personal or family history of tics and/or other abnormal movements as tics can be brought on or exacerbated by stimulant medication. Cardiac risk assessment prior to commencement of medications, including risk factors for sudden cardiac death. An ECG must be completed if indicated. In the case of personal/family history of cardiac disease, ECG is recommended prior to starting children on certain antidepressants and stimulant medication. See the 'Guidance Note: Prescribing stimulant medicines' available on the Stimulant medicines Department of Health webpage **Antidepressant medication** In addition to routine monitoring, patients receiving antidepressants should be screened for precipitation or exacerbation of suicidality. Relevant baseline investigations must be undertaken. Cardiac risk factors must be identified (when on tricyclics). **Mood Stabilisers** This includes Lithium and anti-epileptics. In addition to routine monitoring, renal and thyroid function, and lithium levels must be monitored.	The CAMHS SSCD Physical Examination Form (printable and writeable). Screening Tool for Identification of Potential Cardiac Arrest, MR816.05 (available printed) Screening tool for the identification of potential cardiac risk factors for sudden death among children starting stimulant medication. ECGs can be arranged in the private sector or the Cardiology Department at PCH. See WA Cardiology website for doctors . PCH Cardiologists are available to assist with interpretation of ECGs, phone: 08 6456 0204 or email: pch.cardiologyadmin@health.wa.gov.au or pchoutpatients.cardiology@health.wa.gov.au

Monitoring	
All children receiving psychotropic medication require regular monitoring for side effects of such medication. This includes: height, weight BMI Waist circumference Waist reference intervals Children at risk for metabolic side effects should have their BMI calculated and recorded on the Antipsychotic Physical Health Monitoring Form three-month intervals and their BMI plotted on growth charts.	The Antipsychotic Physical Health Monitoring Form records investigations and monitoring including height, weight, blood pressure and pulse. Girls Stature for Age and Weight for Age Percentiles Boys Stature for Age and Weight for Age Percentiles Girls Body Mass Index-for-Age Percentiles Boys Body Mass Index-for-Age Percentiles Waist reference intervals[3] (See appendix 1)
All children should have baseline and three-monthly measurements of pulse and blood pressure (BP).	Girls Blood Pressure Reference Intervals[2] (See appendix 2) Boys Blood Pressure Reference Intervals[4] (See appendix 2) Pulse reference intervals[5] (See appendix 3)
Agitation and Arousal	Agitation and Arousal medication chart MR816.06 (available printed)
Cardiac Arrest	Screening tool for identification of Potential Cardiac Arrest MR816.05 (available printed)
The date of medication initiation and date and rationale for alterations in medication prescription should be recorded and filed in the medical record. A review of all new medications, including non-psychotropic medications and	Medication Management Plan (inpatient units) MR859.95 (available printed) and guide.

[3] Fernandez JR, Redden DT. Pietrobelli MD, Allison DB. Waist circumference percentiles in nationally representative samples of Afrian-American European-American and Mexican-American children and adolescents. The Journal of Pediatrics. 2004 Oct; 439-44.

[4] U.S. Department of Health and Human Services, National Institutes of Health National Heart, Lung, and Blood Institute. The fourth report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents. U.S. Department of Health and Human Services; Originally printed Sept 1996 NIH Publication No. 05-5267, Revised May 2005.

[5] Kliegman RM, Stanton, BMD, St Geme, S, Schor, NF. Nelson Textbook of Pediatrics. 20th ed. [Internet]. Elsevier; 2016 [cited 2018 Aug 27]. Available from ClinicalKey.

supplements prescribed by GP and medical specialists, in order to avoid potential drug-drug interactions. Discuss outcomes of medication reviews with the MDT as clinically indicated.	CAMHS Multidisciplinary Team Clinical Reviews Policy.
Monitoring: additional requirements for specific medications	
Antipsychotic medication Monitoring of cardiometabolic health Monitoring for Extrapyramidal Side Effects (EPSE), guided by the Antipsychotic Physical Health Monitoring Form. and documented in the CAMHS physical examination. If EPSE are present, it is recommended that Abnormal Involuntary Movement Scale tool is completed Monitoring for hyperprolactinaemia. All children (prepubertal and postpubertal) require baseline measurement of prolactin and ongoing screening for signs of hyperprolactinaemia. It is recommended that prepubertal children on antipsychotic medication require 6 monthly prolactin levels as some screening questions in The Antipsychotic Physical Health Monitoring Form are not appropriate for prepubertal children (questions relevant to them concern blunted growth or delayed puberty). In post pubertal children the presence of symptoms identified in the questionnaire should trigger repeat Prolactin level.	BP and pulse on the Antipsychotic Physical Health Monitoring Form and algorithm. Questions asked to monitor for Extrapyramidal side effects. These questions support screening and ongoing monitoring of EPSE. Abnormal Involuntary Movement Scale Questions asked to monitor for Hyperprolactinemia Clinical Algorithm for Monitoring Metabolic Syndrome in Children
Stimulant medication Screening for abnormal movements, including tics.	Stimulant medication checklist Screening Tool for Identification of Potential Cardiac Arrest, MR816.05 (available printed)
Antidepressants Screening for precipitation or exacerbation of suicidality.	Monitoring Recommendations for

Screened and monitoring for cardiac risk factors must be completed Reviews must be completed annually.	Patients Treated for Major Depressive Disorder.[6]
Actions in the event of abnormal findings	
Clinicians should refer to the Clinical Algorithm for Monitoring Metabolic Syndrome in Children which is a brief guide to support clinical decision making in the event of abnormal findings on physical examination (BMI, BP) or investigations (Fasting, BSL, insulin, cholesterol, and prolactin). This has been developed in consultation with Paediatric Endocrinology at PCH. See the PCH Endocrinology and Diabetes Department intranet page for contact details for the Paediatric Endocrinologist who is available for further advice/consultation.	

[6]Dodd, S. et al. (2001) A consensus statement for safety monitoring guidelines of treatments for major depressive disorder, Australia and New Zealand Journal of Psychiatry, 45(9): 712–725. Accessed from: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3190838/

Policies

CAHS Policies

CAHS Allergy and Adverse Drug Reaction Management policy

CAHS Use of medications for other than Manufacturer's approved indications policy

CAHS High Risk Medicines policy

PCH Lithium monograph

CAMHS Policies

CAMHS Psychotropic Medication Monitoring Adverse Physical Health Effects policy

CAMHS Physical Health Assessment and Monitoring Policy

CAMHS Consent procedure

WA Health Policies

WA Health Medication Chart policy

WA Health Mental health consumer medication information policy

Prescribing information

CAHS Pharmacy Intranet Page

Chief Psychiatrist's Standards for Clinical Care As required under Section 547 of the MHA 2014 See page 15 re metabolic screening.

MIMS online- see CAHS Library and Information Service for link

Stimulant medicines WA Health webpage

Resources

CAMHS Medication Safety Toolbox

Australian Commission on Safety and Quality in Health Care: Resources for medication reconciliation

National Harm Education Course - Clozapine

CAMHS Consent for Medication Use MR859.01/MR505.00*

CAMHS Antipsychotic Physical Health Monitoring Form MR505.01*

* Order from Print Media Group or print from link. Forms must be printed on paper that is not less than 80 grams per square metre (gsm) as per the Australian Standard for Health Records, Part 1: Paper Health Records, in AS 2828.1 (2019).

References

Abnormal Involuntary Movement Scale (available on HealthPoint)

Centres for Disease Control and Prevention – Growth Charts

Department of Health Government of Western Australia [Internet]. Guidance Note: Prescribing stimulant medicine. Department of Health; 2017 [updated 2017, cited 2018 Aug 28]. Available from: https://ww2.health.wa.gov.au/Articles/S_T/Stimulant-medicines

Dodd, S. *et al*. (2001) A consensus statement for safety monitoring guidelines of treatments for major depressive disorder, Australia and New Zealand Journal of Psychiatry, 45(9): 712–725. Accessed from: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3190838/

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Hamilton, R. *et al.* (2009) Cardiac risk assessment before the use of stimulant medications in children and youth: A joint position statement by the Canadian Paediatric Society, the Canadian Cardiovascular Society and the Canadian Academy of Child and Adolescent Psychiatry, Journal of the Canadian Academy of Child and Adolescent Psychiatry, 18(4): 349–355. Accessed from: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2765391/

International Diabetes Federation (IDF) - Definition of metabolic syndrome

Kliegman RM, Stanton, BMD, St Geme, S, Schor, NF. Nelson Textbook of Pediatrics. 20th ed. [Internet]. Elsevier; 2016 [cited 2018 Aug 27]. Available from ClinicalKey.

National Standards for Mental Health Services (NSMHS) - Standard 2

Royal Australian and New Zealand College of Psychiatrists - Clinical guidance on the use of antidepressant medications in children and adolescents March 2005

Shah, A. a, Aftab, A., & Coverdale, J. (2014). QTc prolongation with antipsychotics: is routine ECG monitoring recommended? Journal of Psychiatric Practice, 20(3), 196–206.

WA Therapeutic Advisory Group – WA Psychotropic Drugs Committee WA TAG Clinical Guidelines web page

U.S. Department of Health and Human Services, National Institutes of Health National Heart, Lung, and Blood Institute. The fourth report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents. U.S. Department of Health and Human Services; Originally printed Sept 1996 NIH Publication No. 05-5267, Revised May 2005.

WA Psychotropic Drugs Committee – Anxiety Disorders: Drug Treatment Guidelines 2008

WA Health Public Health Division - Stimulant Medicines - WA Health site

Consent to Treatment Policy for the Western Australian Health System 2016

Women's and Children's Hospital, Government of South Australia - WCHN Antipsychotic physical health and adverse effect monitoring package and Clinical Procedure For Antipsychotic Medication - Monitoring Adverse Effects When Prescribed For Children And Adolescents Prescribed Antipsychotic Medication.

Appendix 1 Waist reference intervals

Table IV. Estimated value for percentile regression for all children and adolescents combined, according to sex

	Percentile for boys					Percentile for girls				
	10th	25th	50th	75th	90th	10th	25th	50th	75th	90th
Intercept	39.7	41.3	43.0	43.6	44.0	40.7	41.7	43.2	44.7	46.1
Slope	1.7	1.9	2.0	2.6	3.4	1.6	1.7	2.0	2.4	3.1
Age (y)										
2	43.2	45.0	47.1	48.8	50.8	43.8	45.0	47.1	49.5	52.2
3	44.9	46.9	49.1	51.3	54.2	45.4	46.7	49.1	51.9	55.3
4	46.6	48.7	51.1	53.9	57.6	46.9	48.4	51.1	54.3	58.3
5	48.4	50.6	53.2	56.4	61.0	48.5	50.1	53.0	56.7	61.4
6	50.1	52.4	55.2	59.0	64.4	50.1	51.8	55.0	59.1	64.4
7	51.8	54.3	57.2	61.5	67.8	51.6	53.5	56.9	61.5	67.5
8	53.5	56.1	59.3	64.1	71.2	53.2	55.2	58.9	63.9	70.5
9	55.3	58.0	61.3	66.6	74.6	54.8	56.9	60.8	66.3	73.6
10	57.0	59.8	63.3	69.2	78.0	56.3	58.6	62.8	68.7	76.6
11	58.7	61.7	65.4	71.7	81.4	57.9	60.3	64.8	71.1	79.7
12	60.5	63.5	67.4	74.3	84.8	59.5	62.0	66.7	73.5	82.7
13	62.2	65.4	69.5	76.8	88.2	61.0	63.7	68.7	75.9	85.8
14	63.9	67.2	71.5	79.4	91.6	62.6	65.4	70.6	78.3	88.8
15	65.6	69.1	73.5	81.9	95.0	64.2	67.1	72.6	80.7	91.9
16	67.4	70.9	75.6	84.5	98.4	65.7	68.8	74.6	83.1	94.9
17	69.1	72.8	77.6	87.0	101.8	67.3	70.5	76.5	85.5	98.0
18	70.8	74.6	79.6	89.6	105.2	68.9	72.2	78.5	87.9	101.0

Fernandez JR, Redden DT. Pietrobelli MD, Allison DB. Waist circumference percentiles in nationally representative samples of African-American European-American and Mexican-American children and adolescents. The Journal of Pediatrics. 2004 Oct; 439-44

Appendix 2 Girls Blood Pressure Reference Intervals and Boys Blood Pressure Reference Intervals

From: U.S. Department of Health and Human Services, National Institutes of Health National Heart, Lung, and Blood Institute. The fourth report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents. U.S. Department of Health and Human Services; Originally printed Sept 1996 NIH Publication No. 05-5267, Revised May 2005.

Girls:

TABLE 4

Blood Pressure Levels for Girls by Age and Height Percentile*

Age (Year)	BP Percentile ↓	Systolic BP (mmHg) ← Percentile of Height →							Diastolic BP (mmHg) ← Percentile of Height →						
		5th	10th	25th	50th	75th	90th	95th	5th	10th	25th	50th	75th	90th	95th
1	50th	83	84	85	86	88	89	90	38	39	39	40	41	41	42
	90th	97	97	98	100	101	102	103	52	53	53	54	55	55	56
	95th	100	101	102	104	105	106	107	56	57	57	58	59	59	60
	99th	108	108	109	111	112	113	114	64	64	65	65	66	67	67
2	50th	85	85	87	88	89	91	91	43	44	44	45	46	46	47
	90th	98	99	100	101	103	104	105	57	58	58	59	60	61	61
	95th	102	103	104	105	107	108	109	61	62	62	63	64	65	65
	99th	109	110	111	112	114	115	116	69	69	70	70	71	72	72
3	50th	86	87	88	89	91	92	93	47	48	48	49	50	50	51
	90th	100	100	102	103	104	106	106	61	62	62	63	64	64	65
	95th	104	104	105	107	108	109	110	65	66	66	67	68	68	69
	99th	111	111	113	114	115	116	117	73	73	74	74	75	76	76
4	50th	88	88	90	91	92	94	94	50	50	51	52	52	53	54
	90th	101	102	103	104	106	107	108	64	64	65	66	67	67	68
	95th	105	106	107	108	110	111	112	68	68	69	70	71	71	72
	99th	112	113	114	115	117	118	119	76	76	76	77	78	79	79
5	50th	89	90	91	93	94	95	96	52	53	53	54	55	55	56
	90th	103	103	105	106	107	109	109	66	67	67	68	69	69	70
	95th	107	107	108	110	111	112	113	70	71	71	72	73	73	74
	99th	114	114	116	117	118	120	120	78	78	79	79	80	81	81
6	50th	91	92	93	94	96	97	98	54	54	55	56	56	57	58
	90th	104	105	106	108	109	110	111	68	68	69	70	70	71	72
	95th	108	109	110	111	113	114	115	72	72	73	74	74	75	76
	99th	115	116	117	119	120	121	122	80	80	80	81	82	83	83
7	50th	93	93	95	96	97	99	99	55	56	56	57	58	58	59
	90th	106	107	108	109	111	112	113	69	70	70	71	72	72	73
	95th	110	111	112	113	115	116	116	73	74	74	75	76	76	77
	99th	117	118	119	120	122	123	124	81	81	82	82	83	84	84
8	50th	95	95	96	98	99	100	101	57	57	57	58	59	60	60
	90th	108	109	110	111	113	114	114	71	71	71	72	73	74	74
	95th	112	112	114	115	116	118	118	75	75	75	76	77	78	78
	99th	119	120	121	122	123	125	125	82	82	83	83	84	85	86
9	50th	96	97	98	100	101	102	103	58	58	58	59	60	61	61
	90th	110	110	112	113	114	116	116	72	72	72	73	74	75	75
	95th	114	114	115	117	118	119	120	76	76	76	77	78	79	79
	99th	121	121	123	124	125	127	127	83	83	84	84	85	86	87
10	50th	98	99	100	102	103	104	105	59	59	59	60	61	62	62
	90th	112	112	114	115	116	118	118	73	73	73	74	75	76	76
	95th	116	116	117	119	120	121	122	77	77	77	78	79	80	80
	99th	123	123	125	126	127	129	129	84	84	85	86	86	87	88

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Girls continued

Age (Year)	BP Percentile ↓	Systolic BP (mmHg) ← Percentile of Height →							Diastolic BP (mmHg) ← Percentile of Height →						
		5th	10th	25th	50th	75th	90th	95th	5th	10th	25th	50th	75th	90th	95th
11	50th	100	101	102	103	105	106	107	60	60	60	61	62	63	63
	90th	114	114	116	117	118	119	120	74	74	74	75	76	77	77
	95th	118	118	119	121	122	123	124	78	78	78	79	80	81	81
	99th	125	125	126	128	129	130	131	85	85	86	87	87	88	89
12	50th	102	103	104	105	107	108	109	61	61	61	62	63	64	64
	90th	116	116	117	119	120	121	122	75	75	75	76	77	78	78
	95th	119	120	121	123	124	125	126	79	79	79	80	81	82	82
	99th	127	127	128	130	131	132	133	86	86	87	88	88	89	90
13	50th	104	105	106	107	109	110	110	62	62	62	63	64	65	65
	90th	117	118	119	121	122	123	124	76	76	76	77	78	79	79
	95th	121	122	123	124	126	127	128	80	80	80	81	82	83	83
	99th	128	129	130	132	133	134	135	87	87	88	89	89	90	91
14	50th	106	106	107	109	110	111	112	63	63	63	64	65	65	66
	90th	119	120	121	122	124	125	125	77	77	77	78	79	80	80
	95th	123	123	125	126	127	129	129	81	81	81	82	83	84	84
	99th	130	131	132	133	135	136	136	88	88	89	90	90	91	92
15	50th	107	108	109	110	111	113	113	64	64	64	65	66	67	67
	90th	120	121	122	123	125	126	127	78	78	78	79	80	81	81
	95th	124	125	126	127	129	130	131	82	82	82	83	84	85	85
	99th	131	132	133	134	136	137	138	89	89	90	91	91	92	93
16	50th	108	108	110	111	112	114	114	64	64	65	66	66	67	68
	90th	121	122	123	124	126	127	128	78	78	79	80	81	81	82
	95th	125	126	127	128	130	131	132	82	82	83	84	85	85	86
	99th	132	133	134	135	137	138	139	90	90	90	91	92	93	93
17	50th	108	109	110	111	113	114	115	64	65	65	66	67	67	68
	90th	122	122	123	125	126	127	128	78	79	79	80	81	81	82
	95th	125	126	127	129	130	131	132	82	83	83	84	85	85	86
	99th	133	133	134	136	137	138	139	90	90	91	91	92	93	93

BP, blood pressure

* The 90th percentile is 1.28 SD, 95th percentile is 1.645 SD, and the 99th percentile is 2.326 SD over the mean. For research purposes, the standard deviations in appendix table B–1 allow one to compute BP Z-scores and percentiles for girls with height percentiles given in table 4 (i.e., the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles). These height percentiles must be converted to height Z-scores given by (5% = -1.645; 10% = -1.28; 25% = -0.68; 50% = 0; 75% = 0.68; 90% = 1.28; 95% = 1.645) and then computed according to the methodology in steps 2–4 described in appendix B. For children with height percentiles other than these, follow steps 1–4 as described in appendix B.

Boys:

TABLE 3

Blood Pressure Levels for Boys by Age and Height Percentile*

Age (Year)	BP Percentile ↓	Systolic BP (mmHg) ← Percentile of Height →							Diastolic BP (mmHg) ← Percentile of Height →						
		5th	10th	25th	50th	75th	90th	95th	5th	10th	25th	50th	75th	90th	95th
1	50th	80	81	83	85	87	88	89	34	35	36	37	38	39	39
	90th	94	95	97	99	100	102	103	49	50	51	52	53	53	54
	95th	98	99	101	103	104	106	106	54	54	55	56	57	58	58
	99th	105	106	108	110	112	113	114	61	62	63	64	65	66	66
2	50th	84	85	87	88	90	92	92	39	40	41	42	43	44	44
	90th	97	99	100	102	104	105	106	54	55	56	57	58	58	59
	95th	101	102	104	106	108	109	110	59	59	60	61	62	63	63
	99th	109	110	111	113	115	117	117	66	67	68	69	70	71	71
3	50th	86	87	89	91	93	94	95	44	44	45	46	47	48	48
	90th	100	101	103	105	107	108	109	59	59	60	61	62	63	63
	95th	104	105	107	109	110	112	113	63	63	64	65	66	67	67
	99th	111	112	114	116	118	119	120	71	71	72	73	74	75	75
4	50th	88	89	91	93	95	96	97	47	48	49	50	51	51	52
	90th	102	103	105	107	109	110	111	62	63	64	65	66	66	67
	95th	106	107	109	111	112	114	115	66	67	68	69	70	71	71
	99th	113	114	116	118	120	121	122	74	75	76	77	78	78	79
5	50th	90	91	93	95	96	98	98	50	51	52	53	54	55	55
	90th	104	105	106	108	110	111	112	65	66	67	68	69	69	70
	95th	108	109	110	112	114	115	116	69	70	71	72	73	74	74
	99th	115	116	118	120	121	123	123	77	78	79	80	81	81	82
6	50th	91	92	94	96	98	99	100	53	53	54	55	56	57	57
	90th	105	106	108	110	111	113	113	68	68	69	70	71	72	72
	95th	109	110	112	114	115	117	117	72	72	73	74	75	76	76
	99th	116	117	119	121	123	124	125	80	80	81	82	83	84	84
7	50th	92	94	95	97	99	100	101	55	55	56	57	58	59	59
	90th	106	107	109	111	113	114	115	70	70	71	72	73	74	74
	95th	110	111	113	115	117	118	119	74	74	75	76	77	78	78
	99th	117	118	120	122	124	125	126	82	82	83	84	85	86	86
8	50th	94	95	97	99	100	102	102	56	57	58	59	60	60	61
	90th	107	109	110	112	114	115	116	71	72	72	73	74	75	76
	95th	111	112	114	116	118	119	120	75	76	77	78	79	79	80
	99th	119	120	122	123	125	127	127	83	84	85	86	87	87	88
9	50th	95	96	98	100	102	103	104	57	58	59	60	61	61	62
	90th	109	110	112	114	115	117	118	72	73	74	75	76	76	77
	95th	113	114	116	118	119	121	121	76	77	78	79	80	81	81
	99th	120	121	123	125	127	128	129	84	85	86	87	88	88	89
10	50th	97	98	100	102	103	105	106	58	59	60	61	61	62	63
	90th	111	112	114	115	117	119	119	73	73	74	75	76	77	78
	95th	115	116	117	119	121	122	123	77	78	79	80	81	81	82
	99th	122	123	125	127	128	130	130	85	86	86	88	88	89	90

Boys continued

Age (Year)	BP Percentile ↓	Systolic BP (mmHg) ← Percentile of Height →							Diastolic BP (mmHg) ← Percentile of Height →						
		5th	10th	25th	50th	75th	90th	95th	5th	10th	25th	50th	75th	90th	95th
11	50th	99	100	102	104	106	107	107	59	59	60	61	62	63	63
	90th	113	114	115	117	119	120	121	74	74	75	76	77	78	78
	95th	117	118	119	121	123	124	125	78	78	79	80	81	82	82
	99th	124	125	127	129	130	132	132	86	86	87	88	89	90	90
12	50th	101	102	104	106	108	109	110	59	60	61	62	63	63	64
	90th	115	116	118	120	121	123	123	74	75	75	76	77	78	79
	95th	119	120	122	123	125	127	127	78	79	80	81	82	82	83
	99th	126	127	129	131	133	134	135	86	87	88	89	90	90	91
13	50th	104	105	106	108	110	111	112	60	60	61	62	63	64	64
	90th	117	118	120	122	124	125	126	75	75	76	77	78	79	79
	95th	121	122	124	126	128	129	130	79	79	80	81	82	83	83
	99th	128	130	131	133	135	136	137	87	87	88	89	90	91	91
14	50th	106	107	109	111	113	114	115	60	61	62	63	64	65	65
	90th	120	121	123	125	126	128	128	75	76	77	78	79	79	80
	95th	124	125	127	128	130	132	132	80	80	81	82	83	84	84
	99th	131	132	134	136	138	139	140	87	88	89	90	91	92	92
15	50th	109	110	112	113	115	117	117	61	62	63	64	65	66	66
	90th	122	124	125	127	129	130	131	76	77	78	79	80	80	81
	95th	126	127	129	131	133	134	135	81	81	82	83	84	85	85
	99th	134	135	136	138	140	142	142	88	89	90	91	92	93	93
16	50th	111	112	114	116	118	119	120	63	63	64	65	66	67	67
	90th	125	126	128	130	131	133	134	78	78	79	80	81	82	82
	95th	129	130	132	134	135	137	137	82	83	83	84	85	86	87
	99th	136	137	139	141	143	144	145	90	90	91	92	93	94	94
17	50th	114	115	116	118	120	121	122	65	66	66	67	68	69	70
	90th	127	128	130	132	134	135	135	80	80	81	82	83	84	84
	95th	131	132	134	136	138	139	140	84	85	86	87	87	88	89
	99th	139	140	141	143	145	146	147	92	93	93	94	95	96	97

BP, blood pressure

* The 90th percentile is 1.28 SD, 95th percentile is 1.645 SD, and the 99th percentile is 2.326 SD over the mean. For research purposes, the standard deviations in appendix table B-1 allow one to compute BP Z-scores and percentiles for boys with height percentiles given in table 3 (i.e., the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles). These height percentiles must be converted to height Z-scores given by (5% = -1.645; 10% = -1.28; 25% = -0.68; 50% = 0; 75% = 0.68; 90% = 1.28; 95% = 1.645) and then computed according to the methodology in steps 2–4 described in appendix B. For children with height percentiles other than these, follow steps 1–4 as described in appendix B.

Appendix 3 Pulse reference intervals

Pulse Rates at Rest, Table 422-4, from:

Kliegman RM, Stanton, BMD, St Geme, S, Schor, NF. Nelson Textbook of Pediatrics. 20th ed. [Internet]. Elsevier; 2016 [cited 2018 Aug 27]. Available from ClinicalKey.

AGE	LOWER LIMITS OF NORMAL (beats/min)		AVERAGE (beats/min)		UPPER LIMITS OF NORMAL (beats/min)	
Newborn	70		125		190	
1–11 mo	80		120		160	
2 yr	80		110		130	
4 yr	80		100		120	
6 yr	75		100		115	
8 yr	70		90		110	
10 yr	70		90		110	
	Girls	**Boys**	**Girls**	**Boys**	**Girls**	**Boys**
12 yr	70	65	90	85	110	105
14 yr	65	60	85	80	105	100
16 yr	60	55	80	75	100	95
18 yr	55	50	75	70	95	90

Psychotropic Medication: Physical Health Monitoring Handbook links

To be used for all children

- CAMHS State-wide Standardised Clinical Documentation (SSCD) Physical examination form - writable
- CAMHS SSCD Physical examination form - printable
- CAMHS Consent for Medication Use MR859.01/MR505.00. (use link or order from Print Media Group [PMG])[1]
- Medication history and management plan (CAMHS IPU only), MR859.95 Find printed form on ward (order from PMG)

Monitoring charts, algorithms, recommendations and screening tools

- Monitoring recommendations for patients treated for major depressive disorder
- Clinical algorithm for monitoring metabolic syndrome in children
- Antipsychotic physical Health monitoring form MR505.01 (use link or order from PMG)[1]
- Screening tool for the identification of potential cardiac risk factors for sudden death among children starting stimulant medication
- Monitoring recommendations for patients treated for major depressive disorders
- WACardiology for doctors
- Agitation and arousal medication chart, MR816.06, Find printed form on ward (order from PMG)
- Screening tool for Identification of Potential Cardiac Arrest, MR816.05 Find printed form on ward (order from PMG)

Specific measurement forms

Growth

- Girls stature for age and weight for age percentiles
- Girls BMI for age percentiles
- Boys stature for age and weight for age percentiles
- Boys BMI for age percentiles
- Waist reference intervals, see appendix 1

Heart rate and blood pressure

- Girls blood pressure intervals, see appendix 3
- Pulse reference intervals, see appendix 4
- Boys blood pressure intervals, see appendix 3

Extrapyramidal side effects

- Abnormal Involuntary Movement Scale

[1] Must be printed on paper that is not less than 80 g per square metre (gsm) as per the Australian Standard for Health Records, Part 1: Paper Health Records, in AS 2828.1 (2019)